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Xingda Lyu

Education

Expected Jun 2027 **B.S. in Statistics, Data Science Track**, *University of Washington*, Seattle, WA.

Expected Jun 2027 **B.S. in Informatics, with Honors**, *University of Washington*, Seattle, WA.
Minor in Philosophy
Cumulative GPA: 3.90/4.00
Selected coursework: Statistical Machine Learning, Data Programming, Data Structures and Algorithms, Statistical Computing, Databases and Data Modeling, Mathematical Statistics & Probability I–III, Generalized Linear Models, Real Analysis.

Research Experience

Aug 2026 – Present ***Research Assistant, Data Mining Group***, *University of Illinois Urbana-Champaign*.
Lead research on AI for science, focusing on retrieval-augmented generation for retrosynthesis and drug discovery. Build and evaluate LLM/retrieval workflows for chemical reasoning and scientific decision support.

Jun 2026 – Aug 2026 ***Research Intern, Data Mining Group***.

Advisor : **Prof. Jiawei Han**, *Michael Aiken Chair Professor, Siebel School of Computing and Data Science*, UIUC (*Personal Web-page*)

Research Mentor : **Joey Chan**, *Ph.D. Candidate, Siebel School of Computing and Data Science*, UIUC (*Profile*)

Aug 2026 – Present ***Research Assistant, Computational Minds and Machines Lab***, *University of Washington*.
Conduct research on computational models of human-AI collaboration, social cognition, and AI alignment. Study how AI systems can reason about users' beliefs, goals, and preferences to support more adaptive and human-centered interaction.

Advisor : **Prof. Max Kleiman-Weiner**, *Assistant Professor, Marketing & International Business; PI, Computational Minds and Machines Lab*, University of Washington (*Personal Web-page*)

Research Mentor : **Kirandeep Kaur**, *Ph.D. Student, Paul G. Allen School of CSE*, University of Washington

Nov 2025 – Aug 2026 ***Research Assistant, InfoSeeking Lab***, *University of Washington*.
Conducted research on epistemic AI agents for information seeking and human-AI collaboration. Designed benchmarks and model-based evaluations for agents that model users' structural ignorance, navigate knowledge gaps, and build stronger epistemic partnerships under uncertainty.

Advisor : **Prof. Chirag Shah**, *Professor, Information School; Adjunct Professor, Paul G. Allen School of CSE*, University of Washington (*Personal Web-page*)

Research Mentor : **Kirandeep Kaur**, *Ph.D. Student, Paul G. Allen School of CSE*, University of Washington

Apr 2026 – Present ***Research Collaborator, UW NLP Group***, *University of Washington*.
Co-lead research on benchmarking whether LLM assistants understand users' well-being, preferences, and conflicts between short-term requests and long-term goals. Develop evaluation protocols and training-oriented analyses for proactive, personalized assistant behavior.

Collaborator : **Stella Li**, *Ph.D. Student, Paul G. Allen School of CSE*, University of Washington (*Personal Web-page*)

Publications & Workshops

Manuscripts

In Preparation Kaur, Kirandeep; **Lyu, Xingda**; Zhou, Ziyu; Cheng, Myra; Li, Shuyue Stella; Chen, Valerie; Tsvetkov, Yulia; and Kleiman-Weiner, Max. *What Uncertainty Conceals: Evaluating Meta-Ignorance in Proactive AI Assistants.* Manuscript in preparation; planned submission to ACL.

In Preparation **Lyu, Xingda**[*]; Chan, Joey[*]; and Han, Jiawei. *RetroRAG: Reaction-Center-Guided Retrieval-Augmented Generation for Retrosynthesis.* Manuscript in preparation; planned submission to KDD. [*]Equal contribution.

Preprints

Under Review **Lyu, Xingda**; Lu, Honglin; Luo, Xinye; and Yang, Shiqi, 2026. *Beyond Instruction-Driven Editing: Source-Grounded Problem Discovery with User-Governed Repair for Scientific Posters.* arXiv:2609.01813. Under review at ACM IUI. (*preprint*)

arXiv 2026 *Agents' Last Exam.* arXiv:2606.05405, 2026. (*preprint*) **Data contributor; author-list update pending.**

In Conference Proceedings

ICML 2026 Kaur, Kirandeep; **Lyu, Xingda**; and Shah, Chirag, 2026. *Position: Knowing Isn't Understanding: Re-grounding Generative Proactivity with Epistemic and Behavioral Insight.* In Proceedings of the International Conference on Machine Learning (ICML 2026). (*paper*)

CDS 2025 **Lyu, Xingda**; Lyu, Gongfu; Yan, Zitai; and Jiang, Yuxin, 2025. *P-RAG: Prompt-Enhanced Parametric RAG with LoRA and Selective CoT for Biomedical and Multi-Hop QA.* In Proceedings of the International Conference on Computing and Data Science (CONF-CDS 2025). (*paper*)

Workshops

CVPR 2026 **Lyu, Xingda**; Luo, Xinye; Lu, Honglin; and Yang, Shiqi, 2026. *PROS: A Proactive Multimodal Refinement Agent for Scientific Posters.* Workshop paper, Human-Interactive Generation and Editing (HiGen), CVPR 2026.

CHIIR 2026 **Lyu, Xingda**, 2026. *Decision-Making Under Unknown Unknowns in Proactive Agentic Systems.* Lightning talk and poster presentation, ACM CHIIR Workshop on Human-Centered Proactive & Personalized Agents.

ARD 2026 Kaur, Kirandeep; Gupta, Vinayak; **Lyu, Xingda**; Gupta, Aditya; and Shah, Chirag, 2026. *Proactive Agents for Knowledge Gap Navigation.* Poster presentation, Amazon Research Day 2026, AI Co-Scientist. (*poster*)

Professional Experience

Jun 2025 – Aug 2025 ***Data Science Intern***, *Amazon*, Seattle, WA.
Built automated data-quality pipelines over 300K+ transportation records using Python and SQL. Developed routing optimization and anomaly-detection analyses to identify operational inefficiencies, refund gaps, and underperforming carriers.

Nov 2024 – May 2025 ***Research Project Lead, P-RAG***, *Carnegie Mellon University*.
Led a research project on retrieval-augmented biomedical and multi-hop question answering. Integrated retrieval, LoRA-based parametric adaptation, selective chain-of-thought prompting, and data augmentation into an experimental QA pipeline. Advisor: **Prof. David P. Woodruff**.

Jul 2024 – Sep 2024 ***Data Science Intern***, *Baynovation*, San Jose, CA.
Built a refinance decision tool using real-time mortgage rate data. Developed an interactive mortgage-scenario dashboard and delinquency-risk models using GLM, Random Forest, and XGBoost.

Honors & Patents

Award ***iSchool Conference Travel Award***, University of Washington ($1,500), 2026.

Funding ***NSF AI-SDM Workshop Travel Funding***, Carnegie Mellon University ($1,000), 2026.

Honor ***Dean's List***, University of Washington, 2023–2026.

Patent **CNIPA-granted utility patents** for automated veterinary pharmaceutical production systems, co-inventor, 2023.

Patent **Chinese invention patent** for a microbial agent for preventing wheat powdery mildew, co-inventor, 2022.

Skills

Languages / Tools Python, R, SQL, Java, JavaScript, LaTeX; Git/GitHub, Linux, Jupyter Notebook

ML / AI PyTorch, Hugging Face Transformers, scikit-learn, LoRA/PEFT, RAG, LLM fine-tuning, multimodal LLM/VLM systems

Research Information retrieval, human-AI interaction, proactive agents, AI for science, benchmark and evaluation design, ablation studies, statistical modeling, qualitative error analysis